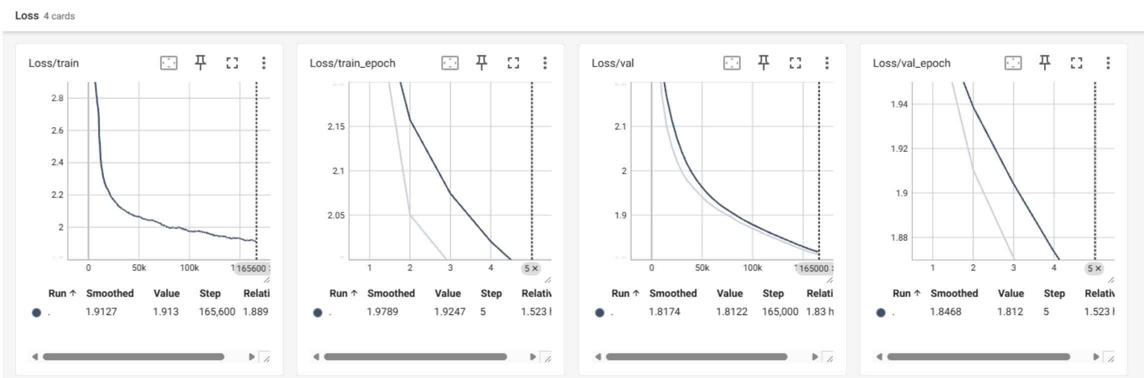
Final Project

Final project report

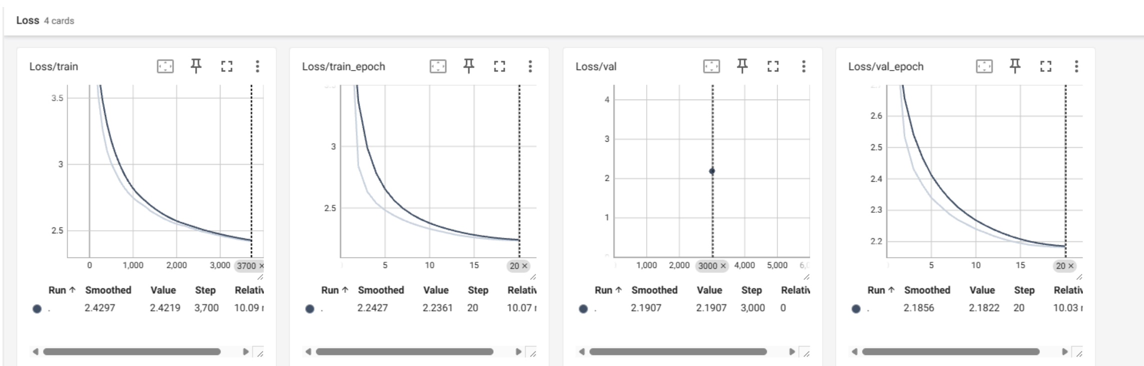
Part 1: Check trainning stability

Foundation model loss



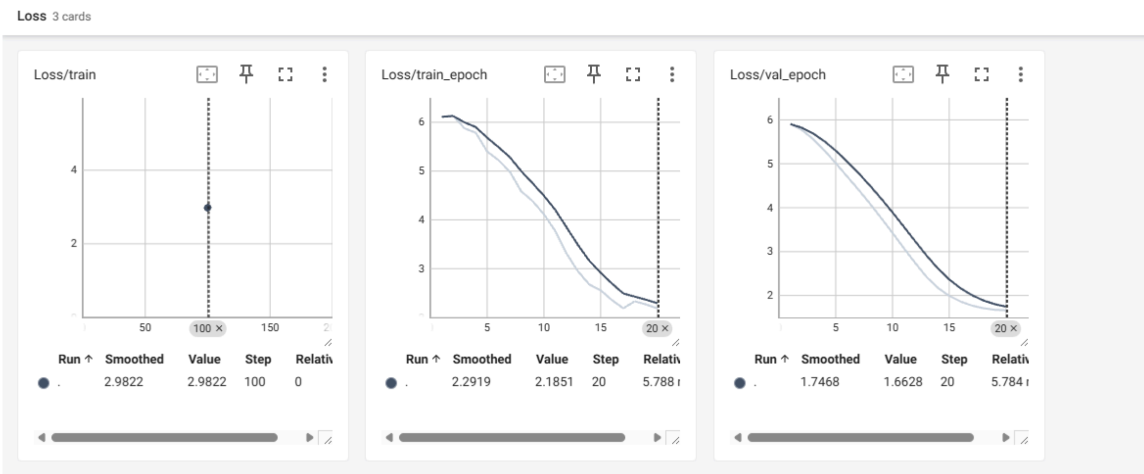
Both training and validation loss decrease smoothly and monotonically throughout training, with no spikes or instability observed. The validation curve closely tracks the training curve without divergence, indicating stable convergence and no evidence of overfitting.

Foundation model instruct tuning loss



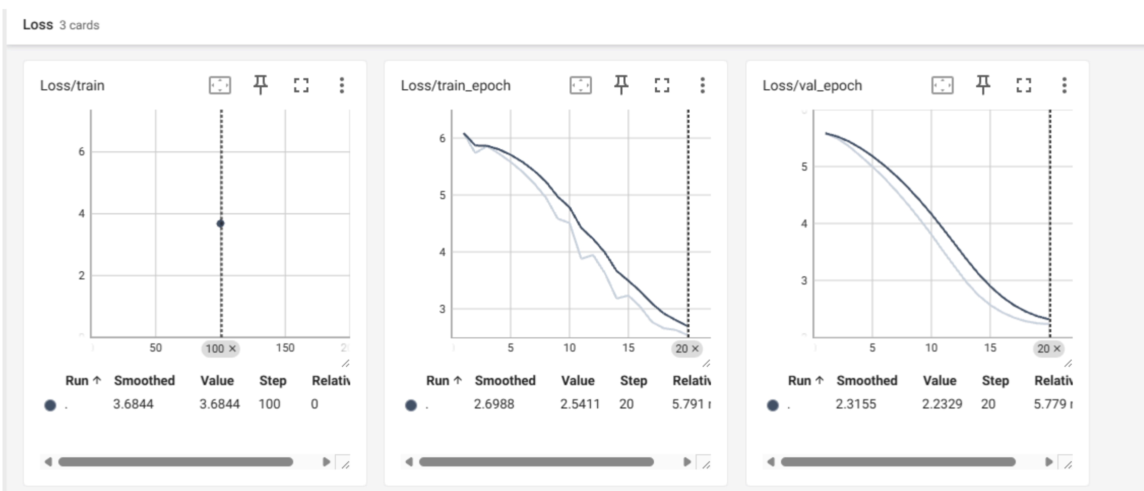
During instruction tuning, the training loss decreases from approximately 3.6 to 2.24, while the validation loss decreases from about 2.7 to 2.18 over 20 epochs. The small and stable gap between final training loss (≈2.24) and validation loss (≈2.18) indicates smooth convergence and no evidence of overfitting.

Pirate dataset instruction tuning



Training loss decreased steadily from approximately 6.1 at the beginning to 2.19 by epoch 20, while validation loss dropped from about 5.9 to 1.66, showing consistent convergence. The validation loss remains slightly lower than training loss at the end (1.66 vs. 2.19), indicating stable generalization with no evidence of overfitting.

Technical writer loss curve



Training loss decreased steadily from approximately 6.0 at the beginning to 2.54 by epoch 20, while validation loss dropped from about 5.8 to 2.23 over the same period. Since validation loss remains close to and slightly lower than training loss at the end (2.23 vs. 2.54), the model shows stable convergence with no clear signs of overfitting.